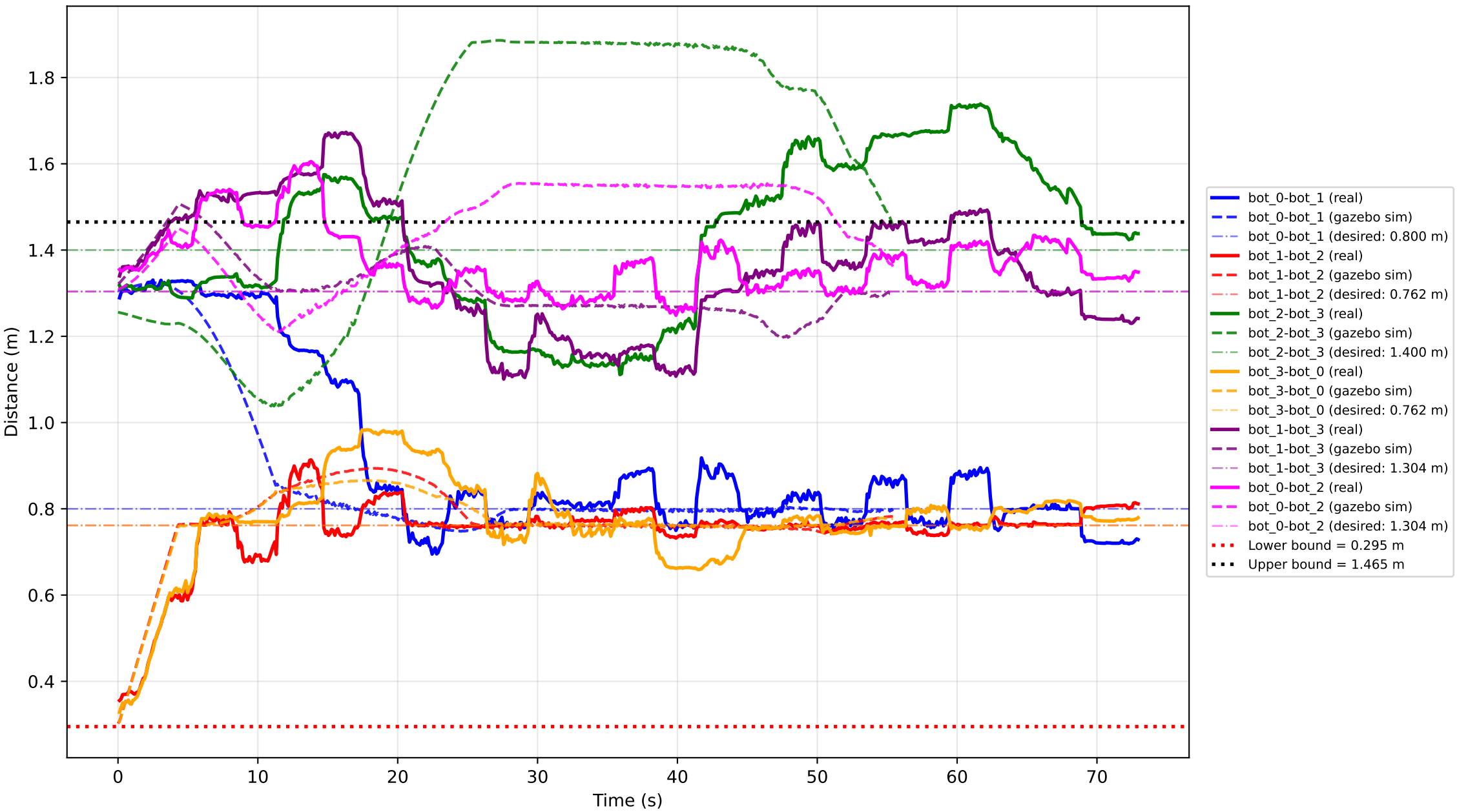

Real (fresh, 2026-10-17) vs. Gazebo Inter-Robot Distances (Case-1b)
Distance (m)
Time (s)
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
0
10
20
30
40
50
60
70
bot_0-bot_1 (real)
bot_0-bot_1 (gazebo sim)
bot_0-bot_1 (desired: 0.800 m)
bot_1-bot_2 (real)
bot_1-bot_2 (gazebo sim)
bot_1-bot_2 (desired: 0.762 m)
bot_2-bot_3 (real)
bot_2-bot_3 (gazebo sim)
bot_2-bot_3 (desired: 1.400 m)
bot_3-bot_0 (real)
bot_3-bot_0 (gazebo sim)
bot_3-bot_0 (desired: 0.762 m)
bot_1-bot_3 (real)
bot_1-bot_3 (gazebo sim)
bot_1-bot_3 (desired: 1.304 m)
bot_0-bot_2 (real)
bot_0-bot_2 (gazebo sim)
bot_0-bot_2 (desired: 1.304 m)
Lower bound = 0.295 m
Upper bound = 1.465 m